



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

JUNE 2011
ASSESSMENT # 3
YEAR 12

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

Total marks—80 Marks

- Attempt all questions.
- The mark-value of each question is boxed in the right margin.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:
Section A (Questions 1 and 2),
Section B (Questions 3 and 4),
Section C (Questions 5 and 6).

Examiner: Mr P. Bigelow

This is an assessment task only and does not necessarily reflect
the content or format of the Higher School Certificate.

Question 1 (12 marks)

- (a) Find a primitive of $3x^2 + 11$. 1
- (b) Express 80° in radians. 1
- (c) Approximate the following correct to three decimal places:
- (i) $\tan 61^\circ 47'$, 1
- (ii) $\cos 3^\circ$, 1
- (iii) $\frac{7}{5e^{1.5}}$. 1
- (d) Find $\int \frac{2x}{7+x^2} dx$. 2
- (e) Express $\sin \frac{7\pi}{3}$ in exact form. 1
- (f) Sketch $y = \cos \pi x$ for $|x| \leq 1$. 2
- (g) Simplify:
- (i) $\ln(e^3)$, 1
- (ii) $e^{4 \ln x}$. 1

Question 2 (14 marks)

(a) Evaluate:

(i) $\int_0^{\frac{\pi}{3}} 3 \sin \frac{x}{2} dx,$

2

(ii) $\int_0^1 e^{4x} dx.$

2

(b) Differentiate, with respect to x :

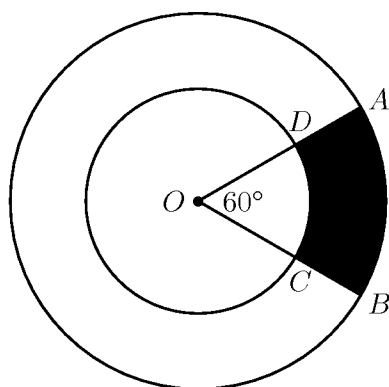
(i) $(e^x - 4)^7,$

2

(ii) $x^2 \cos x.$

2

(c)



The diagram shows two concentric circles, centre O and radii 15 cm and 25 cm. $\widehat{AOB} = 60^\circ$.

Find correct to three significant figures

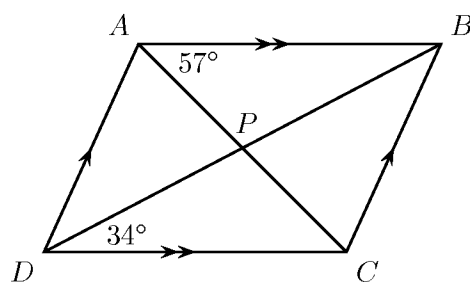
(i) the perimeter of the shaded region $ABCD$,

2

(ii) the area of the shaded region $ABCD$.

2

(d)



2

In the diagram $ABCD$ is a parallelogram whose diagonals intersect at P . Given that $\angle CDB = 34^\circ$ and $\angle BAC = 57^\circ$, find the size of $\angle APB$. Give reasons for your answer.

Section B

(Use a separate writing booklet.)

Marks

Question 3 (13 marks)

- (a) Given the function $y = f(x)$ such that $f'(x) = \frac{2}{\sqrt{x}}$ and $f(4) = 1$,
find the value of $f(9)$.

2

- (b) Find $\frac{d}{dx}(\log_e(x-1))$.

1

- (c) Solve the following equations for x :

(i) $3e^{2x} - e^x = 0$,

2

(ii) $\ln(7x - 12) = 2 \ln x$.

2

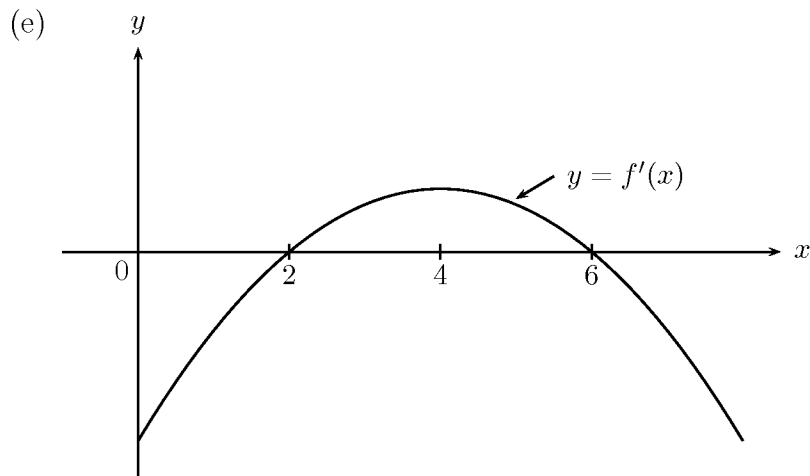
- (d) Given $f(x) = e^{x^3}$,
find:

(i) $f'(x)$,

1

(ii) $f''(x)$.

2



3

The diagram shows the graph of the gradient function of $y = f(x)$.
For what value of x does $y = f(x)$ have a relative maximum?
Justify your answer.

Question 4 (15 marks)

- (a) Find the equation of the tangent to $y = e^{x/2}$ at $P(2, e)$ and show that it passes through the origin.

3

- (b) Consider the curve $y = 2 + 3x - x^3$.

(i) Find $\frac{dy}{dx}$.

1

- (ii) Find the stationary points and determine their nature.

3

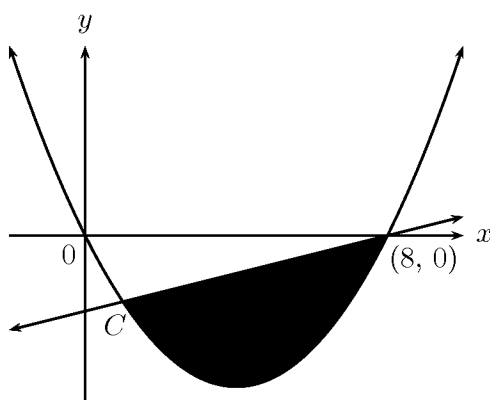
- (iii) For what values of x is the curve concave up?

2

- (iv) Sketch the curve for $-2 \leq x \leq 2$.

2

(c)



The graphs of $y = x - 8$ and $y = x^2 - 8x$ intersect at the points $(8, 0)$ and C .

- (i) Find the co-ordinates of the point C .

2

- (ii) Find the area of the shaded region bounded by $y = x - 8$ and $y = x^2 - 8x$.

2

Section C

(Use a separate writing booklet.)

Marks

Question 5 (12 marks)

(a) For the points $A(2, 7)$ and $B(4, 9)$:

(i) Find the co-ordinates of M , the midpoint of AB .

1

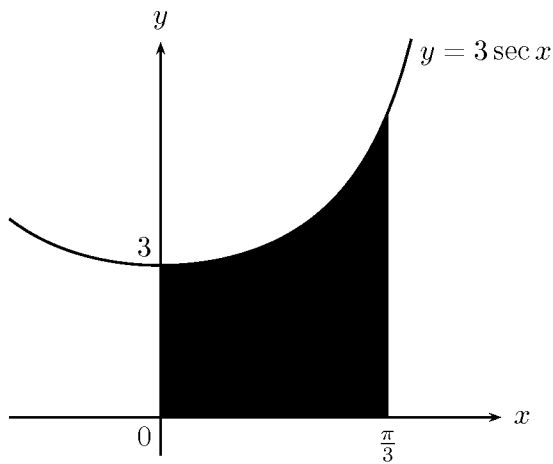
(ii) Show that the equation of the line through M and perpendicular to AB is $x + y - 11 = 0$.

2

(iii) The line in (ii) meets the x -axis at P . Perpendiculars from A and B to the x -axis meet the x -axis at U and V respectively. Prove that the triangles PAU and BPV are congruent.

3

(b)



In the diagram, the shaded region is bounded by the curve $y = 3 \sec x$, the co-ordinate axes, and the line $x = \frac{\pi}{3}$.

The shaded region is rotated about the x -axis.

Calculate the exact volume of the solid.

3

(c) Use calculus to prove that for all rectangles with a perimeter of 36 cm, the maximum area is a square.

3

Question 6 (14 marks)

- (a) (i) Copy and complete the table below with decimals expressed correct to three decimal places. 2

x	3	4	5	6
$\ln x$				

- (ii) Using the table and the Trapezoidal Rule, 2

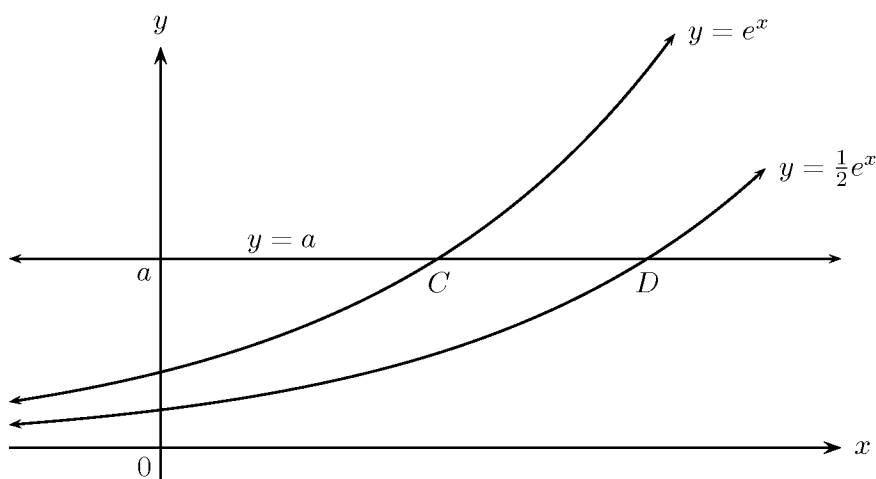
find an approximation to $\int_3^6 \ln x \, dx$.

- (iii) Sketch a graph of $y = \ln x$ and use it to explain why the approximation in (ii) will be less than the exact value of the integral. 2

- (iv) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$. 2

- (v) Deduce the value of the integral in (ii) correct to three decimal places. 2

(b)



A horizontal line, $y = a$ ($a > 1$), is drawn as in the diagram cutting $y = e^x$ and $y = \frac{1}{2}e^x$ at C and D respectively.

- (i) Find the points C and D . 2

- (ii) Show that CD is a constant length. 2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$



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Mathematics Solutions

Question 1 (12 marks)

- (a) Find a primitive of
- $3x^2 + 11$
- .

1

Solution: $x^3 + 11x + c$.

- (b) Express
- 80°
- in radians.

1

Solution: $80^\circ \times \frac{\pi}{180^\circ} = \frac{4\pi}{9}$.

- (c) Approximate the following correct to three decimal places:

- (i)
- $\tan 61^\circ 47'$
- ,

1

Solution: 1.864

- (ii)
- $\cos 3^\circ$
- ,

1

Solution: -0.990

- (iii)
- $\frac{7}{5e^{1.5}}$
- .

1

Solution: 0.312

- (d) Find
- $\int \frac{2x}{7+x^2} dx$
- .

2

Solution: $\ln(7+x^2) + c$.

- (e) Express
- $\sin \frac{7\pi}{3}$
- in exact form.

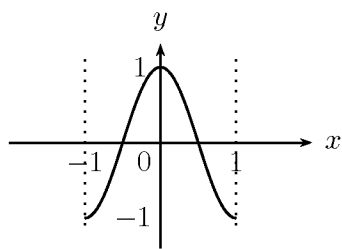
1

Solution: $\sin \frac{7\pi}{3} = \sin \frac{\pi}{3},$
 $= \frac{\sqrt{3}}{2}.$

(f) Sketch $y = \cos \pi x$ for $|x| \leq 1$.

2

Solution:



(g) Simplify:

(i) $\ln(e^3)$,

1

Solution: 3

(ii) $e^{4 \ln x}$.

1

Solution: x^4

Question 2 (14 marks)

(a) Evaluate:

(i) $\int_0^{\frac{\pi}{3}} 3 \sin \frac{x}{2} dx,$

2

Solution: $3 \left[-2 \cos \frac{x}{2} \right]_0^{\frac{\pi}{3}} = -6 \left\{ \cos \frac{\pi}{6} - \cos 0 \right\},$
 $= -6 \left\{ \frac{\sqrt{3}}{2} - 1 \right\},$
 $= 6 - 3\sqrt{3}.$

(ii) $\int_0^1 e^{4x} dx.$

2

Solution: $\left[\frac{1}{4} e^{4x} \right]_0^1 = \frac{1}{4}(e^4 - 1).$

(b) Differentiate, with respect to x :

(i) $(e^x - 4)^7,$

2

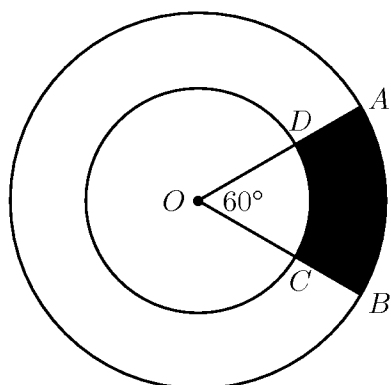
Solution: $7e^x(e^x - 4)^6.$

(ii) $x^2 \cos x.$

2

Solution: $2x \cos x - x^2 \sin x.$

(c)



The diagram shows two concentric circles, centre O and radii 15 cm and 25 cm. $\widehat{AOB} = 60^\circ$.

Find correct to three significant figures

(i) the perimeter of the shaded region $ABCD$,

2

Solution:

$$\begin{aligned}
 AB &= \frac{60}{360} \times 2 \times \pi \times 25, \\
 &= \frac{25\pi}{3}. \\
 DC &= \frac{1}{6} \times 2 \times \pi \times 15, \\
 &= 5\pi. \\
 AD &= 10, \\
 &= BC. \\
 \text{Total} &= \frac{40\pi}{3} + 20, \\
 &\approx 61.9 \text{ cm.}
 \end{aligned}$$

(ii) the area of the shaded region $ABCD$.

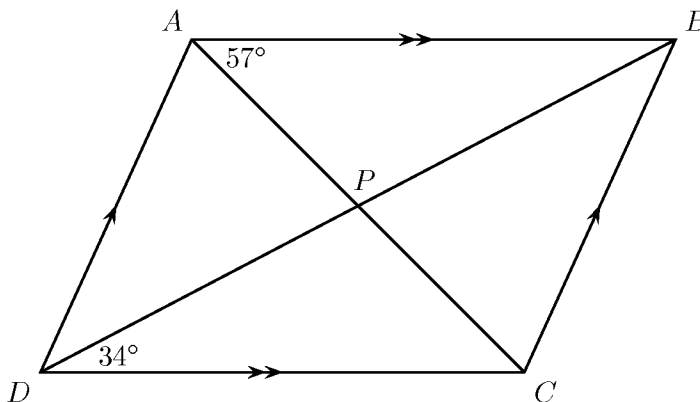
2

Solution:

$$\begin{aligned}
 \text{Sector } AOB &= \frac{1}{6} \times \pi \times 25^2 \text{ or } \frac{1}{2} \times \frac{\pi}{3} \times 25^2, \\
 &= \frac{625\pi}{6}. \\
 \text{Sector } DOC &= \frac{1}{6} \times \pi \times 15^2 \text{ or } \frac{1}{2} \times \frac{\pi}{3} \times 15^2, \\
 &= \frac{225\pi}{6}. \\
 \text{Total} &= \frac{(625 - 225)\pi}{6}, \\
 &\approx 209 \text{ cm}^2.
 \end{aligned}$$

(d)

2



In the diagram $ABCD$ is a parallelogram whose diagonals intersect at P . Given that $\angle CDB = 34^\circ$ and $\angle BAC = 57^\circ$, find the size of $\angle APB$. Give reasons for your answer.

Solution:

$$\begin{aligned}
 \angle CDB &= \angle PBA \text{ (alt. } \angle\text{s, } AB \parallel DC), \\
 180^\circ &= 57^\circ + 34^\circ + \angle APB \text{ (}\angle \text{ sum of } \triangle APB), \\
 \therefore \angle APB &= 89^\circ.
 \end{aligned}$$

Section B

(Use a separate writing booklet.)

Marks

Question 3 (13 marks)

- (a) Given the function $y = f(x)$ such that $f'(x) = \frac{2}{\sqrt{x}}$ and $f(4) = 1$,
find the value of $f(9)$.

2

Solution: $\int 2x^{-\frac{1}{2}} dx = 2 \times 2x^{\frac{1}{2}} + c,$
 $1 = 8 + c \text{ when } x = 4,$
 $\therefore f(x) = 4\sqrt{x} - 7,$
 $f(9) = 5.$

- (b) Find $\frac{d}{dx}(\log_e(x-1))$.

1

Solution: $\frac{1}{x-1}$

- (c) Solve the following equations for x :

(i) $3e^{2x} - e^x = 0,$

2

Solution: $e^x(3e^x - 1) = 0,$
 $3e^x - 1 = 0, \text{ as } e^x \neq 0,$
 $e^x = 1/3,$
 $x = \ln 1/3.$

(ii) $\ln(7x - 12) = 2 \ln x.$

2

Solution: $7x - 12 = x^2,$
 $x^2 - 7x + 12 = 0,$
 $(x - 3)(x - 4) = 0,$
 $\therefore x = 3, 4.$

- (d) Given $f(x) = e^{x^3}$,
find:

(i) $f'(x),$

1

Solution: $3x^2 e^{x^3}$

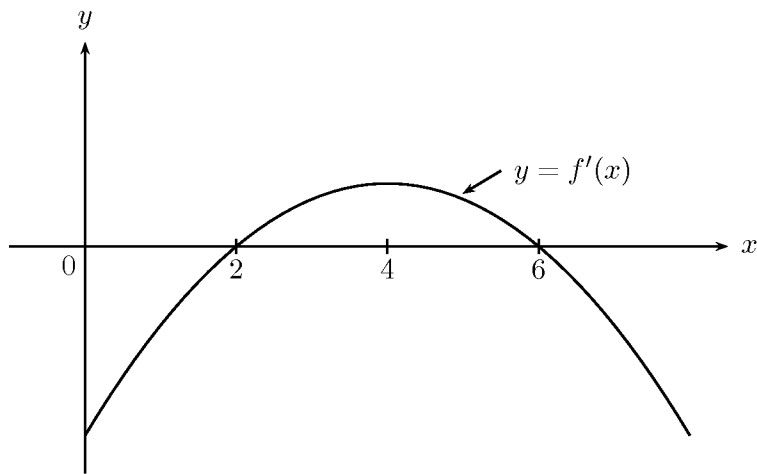
(ii) $f''(x)$.

2

Solution: $3 \times 2x \times e^{x^3} + 3x^2 \times 3x^2 e^{x^3} = e^{x^3}(6x + 9x^4)$.

(e)

3



The diagram shows the graph of the gradient function of $y = f(x)$.
For what value of x does $y = f(x)$ have a relative maximum?
Justify your answer.

Solution: The slope is positive when $2 < x < 6$, zero when $x = 6$, and negative when $x > 6$, so there is a relative maximum at $x = 6$.

Question 4 (15 marks)

- (a) Find the equation of the tangent to $y = e^{x/2}$ at $P(2, e)$ and show that it passes through the origin.

3

Solution:
$$\frac{dy}{dx} = \frac{1}{2}e^{x/2},$$
$$= \frac{e}{2} \text{ at } P(2, e).$$

Tangent: $y - e = \frac{e}{2}(x - 2),$

$$2y - 2e = ex - 2e,$$

$$\text{i.e., } ex - 2y = 0.$$

Substituting $(0, 0)$ gives

$$\text{L.H.S.} = e \times 0 - 2 \times 0,$$

$$= 0,$$

$$= \text{R.H.S.}$$

So the tangent passes through the origin.

- (b) Consider the curve $y = 2 + 3x - x^3$.

- (i) Find $\frac{dy}{dx}$.

1

Solution:
$$\frac{dy}{dx} = 3 - 3x^2.$$

- (ii) Find the stationary points and determine their nature.

3

Solution: When $3 - 3x^2 = 0,$

$$x^2 = 1,$$

$$x = \pm 1.$$

$$\text{If } x = 1, \quad y = 4,$$

$$\text{and if } x = -1, \quad y = 0.$$

$$\frac{d^2y}{dx^2} = -6x,$$

$$= -6 \text{ when } x = 1,$$

$$= 6 \text{ when } x = -1.$$

So there is a maximum at $(1, 4)$ and a minimum at $(-1, 0)$.

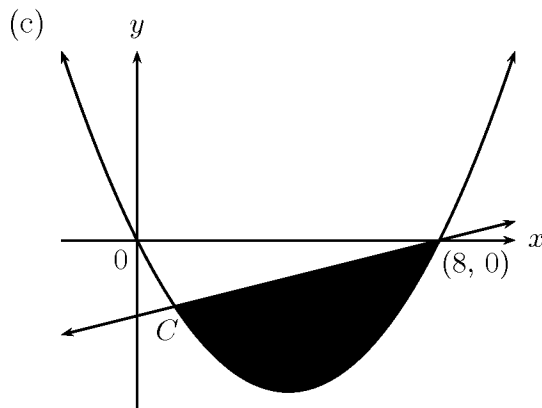
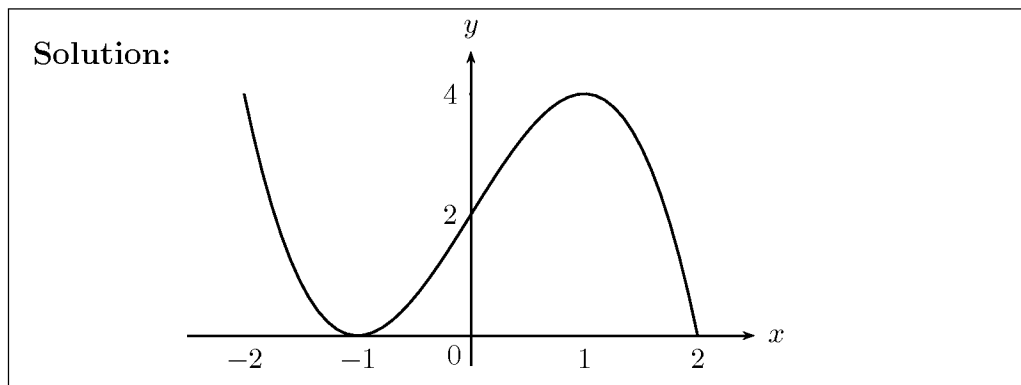
- (iii) For what values of x is the curve concave up?

2

Solution:
$$\frac{d^2y}{dx^2} = -6x > 0,$$
$$\text{i.e., } x < 0.$$

(iv) Sketch the curve for $-2 \leq x \leq 2$.

2



The graphs of $y = x - 8$ and $y = x^2 - 8x$ intersect at the points $(8, 0)$ and C .

(i) Find the co-ordinates of the point C .

2

Solution: $x^2 - 8x = x - 8$,
 $x(x - 8) - (x - 8) = 0$,
 $(x - 1)(x - 8) = 0$,
 $\therefore x = 1, 8$,
 $y = -7, 0$.
 So C is the point $(1, -7)$.

(ii) Find the area of the shaded region bounded by $y = x - 8$ and $y = x^2 - 8x$.

2

Solution: Area = $\int_1^8 ((x - 8) - (x^2 - 8x)) dx$,
 $= \int_1^8 (-x^2 + 9x - 8) dx$,
 $= \left[-\frac{x^3}{3} + \frac{9x^2}{2} - 8x \right]_1^8$,
 $= -\frac{512}{3} + \frac{576}{2} - 64 - \left(-\frac{1}{3} + \frac{9}{2} - 8 \right)$,
 $= \frac{343}{6}$.

Section C

(Use a separate writing booklet.)

Marks

Question 5 (12 marks)

(a) For the points $A(2, 7)$ and $B(4, 9)$:

(i) Find the co-ordinates of M , the midpoint of AB .

1

Solution: $\left(\frac{2+4}{2}, \frac{7+9}{2}\right) = (3, 8)$

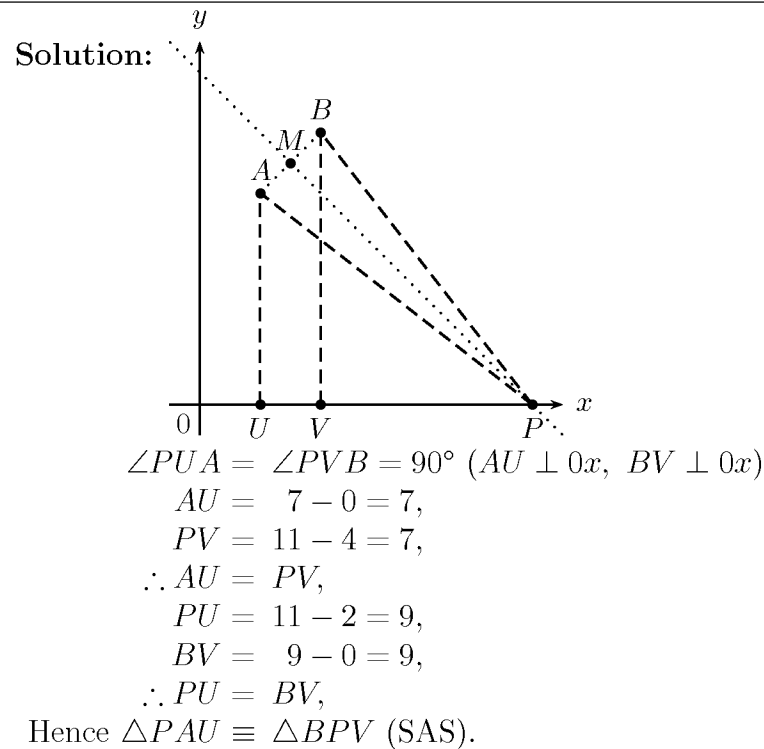
(ii) Show that the equation of the line through M and perpendicular to AB is $x + y - 11 = 0$.

2

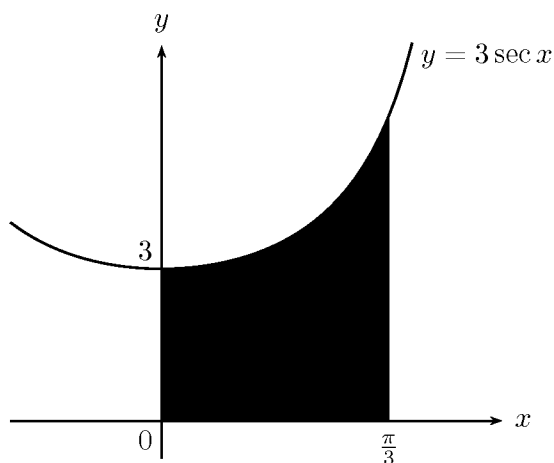
Solution: Slope of $AB = \frac{9-7}{4-2} = 1$.
 $y - 8 = -1 \times (x - 3),$
 $y - 8 = -x + 3,$
 $x + y - 11 = 0.$

(iii) The line in (ii) meets the x -axis at P . Perpendiculars from A and B to the x -axis meet the x -axis at U and V respectively. Prove that the triangles PAU and BPV are congruent.

3



(b)



In the diagram, the shaded region is bounded by the curve $y = 3 \sec x$, the co-ordinate axes, and the line $x = \frac{\pi}{3}$.

The shaded region is rotated about the x -axis.

Calculate the exact volume of the solid.

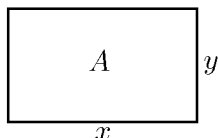
3

Solution: Volume = $\pi \int_0^{\frac{\pi}{3}} 9 \sec^2 x \, dx$,
 $= 9\pi [\tan x]_0^{\frac{\pi}{3}}$,
 $= 9\sqrt{3}\pi$.

- (c) Use calculus to prove that for all rectangles with a perimeter of 36 cm, the maximum area is a square.

3

Solution:



$$2x + 2y = 36,$$

$$y = 18 - x.$$

$$\text{Area, } A = xy,$$

$$= 18x - x^2.$$

$$\frac{dA}{dx} = 18 - 2x,$$

$$= 0 \text{ when } x = 9.$$

$$\frac{d^2A}{dx^2} = -2.$$

$\therefore$ The maximum area occurs when $x = 9$ cm.

i.e., The shape is a square.

Question 6 (14 marks)

- (a) (i) Copy and complete the table below with decimals expressed correct to three decimal places.

2

x	3	4	5	6
$\ln x$				

Solution:

x	3	4	5	6
$\ln x$	1.099	1.386	1.609	1.792

- (ii) Using the table and the Trapezoidal Rule,

2

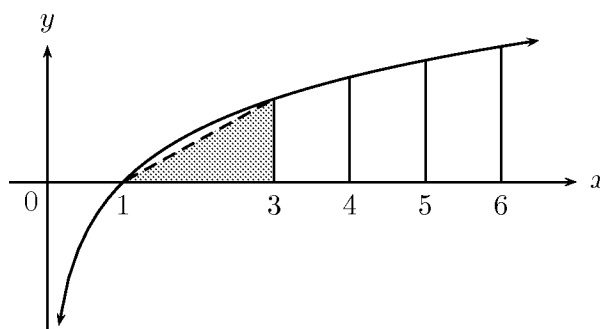
find an approximation to $\int_3^6 \ln x \, dx$.

Solution:
$$\int_3^6 \ln x \, dx \approx \frac{1}{2} \{ 1.099 + 2 \times 1.386 + 2 \times 1.609 + 1.792 \},$$

$$\approx 4.4405 \text{ or about } 4.44.$$

- (iii) Sketch a graph of $y = \ln x$ and use it to explain why the approximation in (ii) will be less than the exact value of the integral.

2

Solution:

As the curve is always concave downwards, any straight line between two points on the curve will always have a small part of the curve area above it (as is shown in exaggerated form between 1 and 3, above). This series of small pieces is not included in the trapeziums between 3 and 4, 4 and 5, and 5 and 6. Therefore the estimated area will be a little less than the real area.

- (iv) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$.

2

Solution:
$$\begin{aligned} \frac{d}{dx}(x \ln x - x) &= \ln x + x \times \frac{1}{x} - 1, \\ &= \ln x + 1 - 1, \\ &= \ln x. \end{aligned}$$

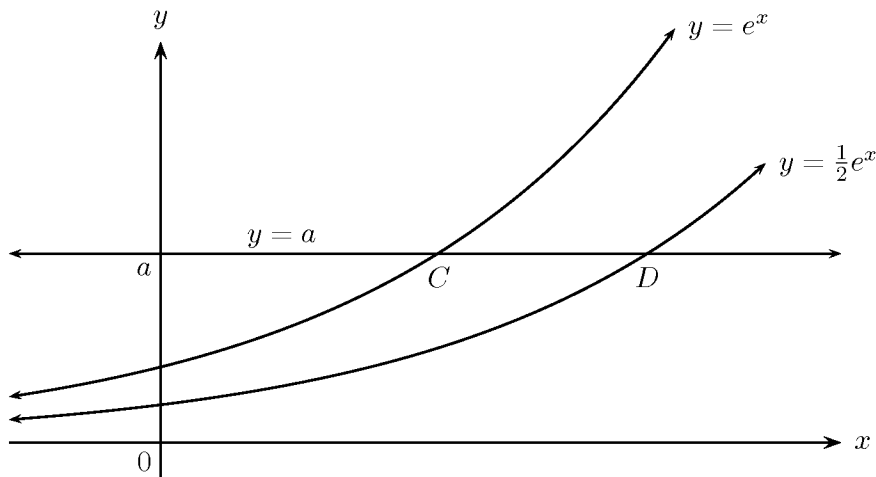
(v) Deduce the value of the integral in (ii) correct to three decimal places.

2

Solution: From (iv) we see that

$$\begin{aligned}\int_3^6 \ln x \, dx &= [x \ln x - x]_3^6, \\ &= 6 \ln 6 - 6 - (3 \ln 3 - 3), \\ &\approx 4.455 \text{ to three decimal places.}\end{aligned}$$

(b)



A horizontal line, $y = a$ ($a > 1$), is drawn as in the diagram cutting $y = e^x$ and $y = \frac{1}{2}e^x$ at C and D respectively.

(i) Find the points C and D .

2

Solution: At C , $a = e^x$,
 $x = \ln a$,
i.e., $C (\ln a, a)$.
 At D , $a = \frac{1}{2}e^x$,
 $x = \ln 2a$,
i.e., $D (\ln 2a, a)$.

(ii) Show that CD is a constant length.

2

Solution: $|CD| = \ln 2a - \ln a$,
 $= \ln \left(\frac{2a}{a} \right)$,
 $= \ln 2$, a constant.